

“032.001.00112 Incubator 1 Temperature is out of range” during Advanced Clean

This is a known interaction between HT Incubator & MW Advance Clean. The error should be suppressed, and there is little risk invalidating test because Advanced Clean occurs outside of assay processing. If the temperature error shows at the GUI interface, this is a defect. If they show up in a Health Report during MW advanced cleaning, this is considered normal. However, if the temperature errors show up at the GUI interface during assay processing invalidate tests.

If we look at the Activity logs, we observe that on 08/07/2012 at 05:00:22AM Maintenance Task 'Mag Wash Clean' started, and ended successfully at 08/07/2012 at 05:25:27AM

Then if we look at the Health Report we observe some Vacuum pressures out of range and are due to the Advanced MagWash Cleaning process where the vacuum is turned off intentionally.

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08/07/2012 05:00:02.407 024.001.00064 64 VACUUMCOOLING Fatal VacuumCooling Vacuum pressure is out of range
08/07/2012 05:19:05.893 024.001.00064 64 VACUUMCOOLING Fatal VacuumCooling Vacuum pressure is out of range
08/07/2012 05:23:43.930 024.001.00064 64 VACUUMCOOLING Fatal VacuumCooling Vacuum pressure is out of range
08/07/2012 05:24:28.543 024.001.00064 64 VACUUMCOOLING Fatal VacuumCooling Vacuum pressure is out of range
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Also, if we take a look at the Message logs, there's an Incubator1 Temperature out of range on 08/07/2012 at 05:18:30AM which is within the time frame when MagWash Advance cleaning is taking place. However, it's just a WARNING and doesn't cause a system failure.

Brief Explanation of Advanced MagWash Cleaning Process.

1. ~15minutes later the Advanced Clean was initiated. During that time 2 MTUs are placed in the Incubator1 (HT incubator).
2. These cold MTUs with ~2.5mL (Endozyme) shocked the HT incubator temperature causing it to drop ~ 0.7°C; thereafter overshooting by ~0.5°C.(See HT Incubator Temperature Graph at the bottom)
3. ~2minutes later the warm MTUs are placed in the MWS1 & 2.
4. Note: Advanced Clean also correctly shuts off vacuum.*

*Most importantly, the error “Vacuum pressure is out of range” messages are correctly **Suppressed** from the **Message logs**, and do not cause a system failure. However, they do show up in Health Report and are considered normal.

Activity_2012-08-06.pdf - Adobe Reader			
File Edit View Document Tools Window Help			
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8/7/2012 8:10:35 AM	smiths	User logged on	
8/7/2012 8:10:09 AM	slingerc	User logged off	
8/7/2012 7:39:41 AM	slingerc	System state entered: Power Save	
8/7/2012 6:39:33 AM	slingerc	System state entered: Ready	
8/7/2012 6:39:33 AM	slingerc	System state entered: Setup	
8/7/2012 6:30:52 AM	slingerc	Fluids priming started	
8/7/2012 6:30:42 AM	slingerc	System state entered: Priming Instrument	
8/7/2012 6:30:03 AM	slingerc	System state entered: Setup	
8/7/2012 6:25:41 AM	slingerc	System state entered: Power Save	
8/7/2012 5:25:27 AM	slingerc	System state entered: Setup	
8/7/2012 5:25:27 AM	slingerc	Maintenance task 'Mag Wash Clean' completed successfully	
8/7/2012 5:00:22 AM	slingerc	System state entered: Maintenance	
8/7/2012 5:00:22 AM	slingerc	Maintenance task 'Mag Wash Clean' started	
8/7/2012 5:00:03 AM	slingerc	System state entered: Setup	
8/6/2012 7:42:34 PM	slingerc	System state entered: Power Save	
8/6/2012 6:41:39 PM	slingerc	System state entered: Ready	
8/6/2012 3:41:31 PM	slingerc	MTUs loaded	
8/6/2012 3:39:09 PM	slingerc	LIS configuration edited	
8/6/2012 3:38:21 PM	slingerc	LIS configuration edited	
8/6/2012 3:07:44 PM	slingerc	System state entered: Assay Processing	
8/6/2012 3:07:44 PM	slingerc	System state entered: Ready	
8/6/2012 3:07:43 PM	slingerc	System state entered: Setup	
8/6/2012 2:57:45 PM	slingerc	Test Order added to Sample 122170372: Assay CT/GC, Analytes 3, Priority No, Count 1	
8/6/2012 2:57:32 PM	slingerc	Test Order added to Sample 122161587: Assay CT/GC, Analytes 3, Priority No, Count 1	
8/6/2012 2:57:23 PM	slingerc	Test Order added to Sample 122170302: Assay CT/GC, Analytes 3, Priority No, Count 1	
8/6/2012 2:57:01 PM	slingerc	Rack Loaded: 004940	
8/6/2012 2:56:38 PM	slingerc	Fluids priming started	
8/6/2012 2:56:30 PM	slingerc	System state entered: Priming Instrument	
8/6/2012 2:56:02 PM	slingerc	System state entered: Setup	

HealthReport.pdf - Adobe Reader							
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		out of range		COOLING			
08/06/2012 06:30:09.323	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/06/2012 07:52:53.743	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/06/2012 08:32:07.180	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/06/2012 08:36:45.210	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/06/2012 08:37:29.850	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/06/2012 14:50:56.650	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/06/2012 14:53:40.013	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 05:00:02.407	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 05:19:05.893	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 05:23:43.930	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 05:24:28.543	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 06:30:02.953	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 08:10:41.973	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 09:31:07.170	024.001.00064	Vacuum pressure is out of range	64	VACUUM COOLING	Fatal	VacuumCooling	
08/07/2012 09:52:30.927	081.000.00093	used OLV Fluid Value from Flash	93	OUTPUT QUEUE	Message	OutputQueue	
08/07/2012 09:52:31.113	262.000.00034	Deactivation Buffer A bottle is empty or not connected. Please check bottle connection	34	InventoryManagement	Critical	InventoryManagement	

Messages_2012-08-07.pdf - Adobe Reader

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PANTHER

MESSAGES LOG REPORT

Serial Number: 00368	System SW Version: 3.2.4.6
Date Range: 8/7/2012 12:00:00 AM to 8/7/2012 11:59:59 PM	
Reviewed By: _____	Verified By: _____
Date: _____	Date: _____

Time	Severity	ID	Message
8/7/2012 9:59:48 AM	Critical	257.000.00069	Prime failed. Check universal fluid bottles and repeat prime.
8/7/2012 9:53:15 AM	Critical	262.000.00035	Deactivation Buffer B bottle is empty or not connected. Please check bottle connection
8/7/2012 9:52:52 AM	Critical	262.000.00032	Wash Buffer A bottle is empty or not connected. Please check bottle connection
8/7/2012 9:52:31 AM	Critical	262.000.00034	Deactivation Buffer A bottle is empty or not connected. Please check bottle connection
8/7/2012 5:18:30 AM	Warning	032.001.00112	Incubator 1 temperature is out of range

